

Individual Project 3: Deadline 3/6

Students must implement **from scratch** (without using C++ standard libraries) **one of** the following data structures:

- Linked list
- Stack
- Queue

This custom data structure must be used to **store the objects** of the classes you designed in **Individual Projects 1 and 2**.

Your `main.cpp` file must clearly demonstrate the **core operations** of your chosen data structure:

- Linked list: Insert, Delete, Traverse, Search, ...
- Stack: Push, Pop, isEmpty,
- Queue: Enqueue, Dequeue, Front, isEmpty, ...

Note: Using your chosen data structure to solve a meaningful, real-world problem within your specific project domain in Individual Project 1, 2 is highly encouraged. Examples of practical implementations:

- **Queue** - *Ticket Booking System*: Implement a Queue to manage a **waiting list** for customers when a flight or movie screening is fully booked.
- **Stack** - *Management System*: Implement a Stack to track **transaction history**, allowing users to **undo** the most recent action (e.g., cancelling the last ticket booked).
- **Linked List** - *E-commerce System*: Use a Linked List to manage a **shopping cart**, allowing users to dynamically add or remove items at any position — without the size limitations of a static array.